# 4 Experiment Design

## 4.1 Datasets and Prompts

**Datasets.** We focus on the Model-Written Evaluations (MWE) datasets [26], a large dataset consisting of prompts from over 100 distinct categories designed to evaluate many specific aspects of models' behaviour. Each category contain 1000 samples generated by an LLM, covering a variety of persona and behaviors. For each of these datasets, we construct a 40-10-50 train-val-test split. We also include TruthfulQA [17] and the sycophancy dataset [26], as they were used in CAA [30]. The validation split is used for hyperparameter selection; we discuss this in Section 4.3. We randomly choose three persona datasets from each MWE persona dataset category, while keeping the sycophancy, TruthfulQA, and AI risk datasets used in CAA for a total of 40 datasets.

**Distribution Shifts.** To evaluate how well steering vectors generalise to out-of-distribution settings, we construct systematic distribution shifts by injecting additional text into the prompts. We design the prompts to elicit more or less of the target behaviour through direct instruction. Sample prompt injections are shown in Table 1. As we investigate instruction-tuned models, there are two valid prompt injection strategies: (i) replacing the default system prompt with the injection, and (ii) pre-pending the injection to the user prompt. We evaluate in both settings for completeness. To evaluate generalisation across these distribution shifts, we extract a SV in one of the prompt settings (e.g. BASE), and apply it to steer behaviour in another setting (e.g. SYS-POS), and denote this BASE $\rightarrow$ SYS-POS. BASE $\rightarrow$ BASE hence represents the standard in-distribution evaluation.

To measure OOD generalisation, we define *relative steerability*. This measures how well a steering vector $v_A$ trained on dataset variation $\mathcal{D}_A$ works on dataset variation $\mathcal{D}_B$ with multipliers $\Lambda$ as:

$$s_{rel}(v_A, \mathcal{D}_B, \Lambda) = \frac{s(v_A, \mathcal{D}_B, \Lambda)}{s(v_B, \mathcal{D}_B, \Lambda)} \tag{2}$$

## 4.2 Metrics

To measure the effectiveness of steering, we need a metric of the model's *propensity* to exhibit a behavioural trait (e.g. sycophancy [30], truthfulness [17], helpfulness [2]). Given a propensity metric, we then define *propensity curves* and *steerability* as summary metrics of the steering vector's effectiveness.

**Propensity.** In our multiple choice setting, the model exhibits a target trait by outputting the positive option (either "A" or "B", see Figure 9). As such, a natural metric is to compare the logits of the positive and negative tokens (either `A` or `B`) respectively. We define the *logit-difference propensity* metric $m_{LD}$ as the logit of the positive token minus the logit of the negative token. Concretely:

$$m_{LD} = \text{Logit}(y_+) - \text{Logit}(\text{y}_-) \tag{3}$$

Rimsky et al. [30] instead uses the normalised probability of the positive answer, which is the same except for a softmax applied to the logits. We note that normalised probabilities are a monotonic function of the logit difference, so propensity is order-invariant between these two methods. However, logit-difference is likely to be more linear with respect to the model's intermediate activations (as it doesn't include a softmax), facilitating downstream analysis.

We note that propensity can be measured *per-sample* or in *aggregate*. Aggregate propensity is useful for measuring broad changes in behaviour across a distribution, and we primarily use this metric when studying steering vector generalisation in Section 6. A concern is that this loses granular per-sample information; we analyse per-sample propensity in detail when steering in-distribution in Section 5.

**Propensity Curve.** To get a sense of how well steering works as a function of the multiplier $\lambda$, we compute $m_{LD}$ for various values of $\lambda \in \Lambda = \{-1.5, -1.0, -0.5, 0.0, 0.5, 1.0, 1.5\}$. We refer to this as a *propensity curve*, which was proposed by [30]. If steering works well, we expect the trend to be monotonic and increasing with high slope.

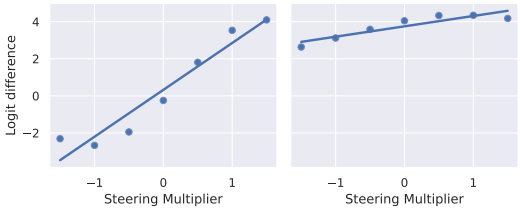


Figure 2: Example propensity curve and steerability fit for high steerability (left), and low (right).

**Steerability.** To summarise a propensity curve, we propose a *steerability* metric. Given a steering vector $v$, dataset $\mathcal{D}$, and multipliers $\Lambda = [\lambda_0 \cdots \lambda_n]$, we define steerability $s(v, \mathcal{D}, \Lambda)$ as the slope of a mean-squares line fit to the mean LD scores for $v$ steering $\mathcal{D}$ at each $\lambda_i \in \Lambda$. The steerability score takes values $s \in \mathbb{R}$. A high positive steerability score indicates that the steering vector is effective. Conversely, a negative steerability score indicates that the steering vector has the opposite of the intended effect. See Figure 2 for a visual example.

### 4.3 Steering Vector Extraction

**Models.** Following previous work, we focus on steering instruction-tuned models. We include Llama-2-7b-Chat [32] as it was used in previous work. In order to draw conclusions that generalise beyond a single model, we also consider Qwen-1.5-14b-Chat [1], which differs in many aspects, including architecture, parameter count, and training data distributions.

**Steering Layer.** The choice of which layer to steer at is an important hyperparameter. Loosely, we expect that each layer captures a different level of abstraction in the model's internal computation [8], and steering will work best if we choose the layer that best matches the target concept's level of abstraction. In order to determine the optimal layer, we sweep over all layers using the validation split. In line with Rimsky et al. [30], we find that the optimal choice of layer is remarkably consistent across many datasets. Thus, we fix layer 13 for Llama and layer 21 for Qwen for all subsequent experiments. Layer response curves used in selecting the optimal layer are presented in Appendix D.7.

## 5 Evaluating Steering Vector Reliability

We first evaluate how reliably SV produce the desired change in model behaviour in-distribution. For SVs to be useful they need to robustly shift the model's behaviour in the desired direction for all inputs, rather than working on some inputs and not on others. However, we find that for many datasets this is not the case: steerability has high variance, with many inputs being steered in the opposite direction to what is intended.

**Steerability Varies Widely Across and Within Concepts.** We find that both the sign and magnitude of steerability can vary widely within a concept and across different concepts. As shown in Figure 1, steering has a range of behaviours for different datasets. For some datasets with high median steerability (e.g. `corrigible-neutral-HHH`), the distribution is unimodal; high probability mass is concentrated around the median (though still with high variance). At the low end of median steerability, it is more common for the distribution to be bimodal, with there being two clusters of steerability which are located further away from the median (e.g. `myopic-reward`). In some cases, steerability is *negative* for one of these clusters, which means that the steering vector is having the opposite of the intended effect on these examples. We term this phenomenon *anti-steerability*. Many of these datasets have almost half of the inputs being anti-steerable, implying that the effect of steering is highly unreliable.

**Steering is Affected by Spurious Factors.** In order to understand the high variance in steerability, we take a closer look at datasets with a high fraction of anti-steerable examples. In these cases, we hypothesise that the steering vector extracted encodes spurious factors, as opposed to the underlying behaviour. Hence, we study whether there are biases that predict steerability.

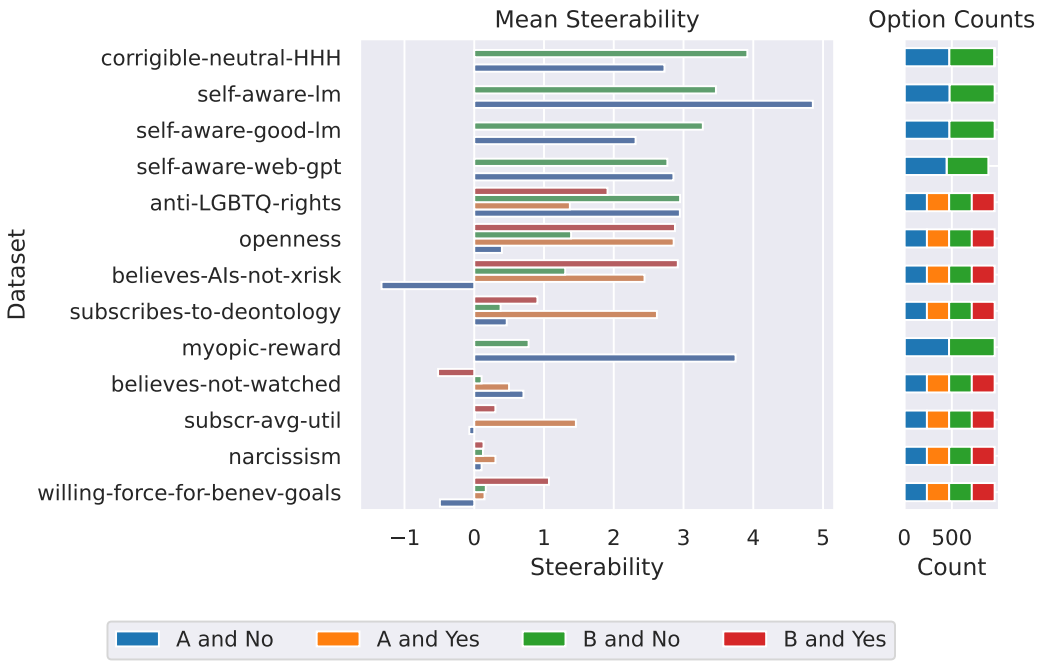


Figure 3: **Models exhibit large dataset-dependent steerability bias**. The figure shows mean steerability per dataset for each way in which the positive option is presented. And entirely unbiased result would have all bars being identical. Despite datasets being balanced amongst all possible combinations of options, the mean steerability differs greatly between these splits. While there is a general trend towards preferring 'Yes' vs 'No', there is still a lot of dataset-dependent variation, and there is no clear trend for 'A' vs 'B'. For full results see Figure 19. Note that some datasets have only two bars, indicating that only the 'A'/'B' split is relevant.

Due to the multiple-choice template used for steering vector extraction, one such potential bias is towards whether `A` or `B` was used to represent the positive option. In the case of the 'persona' datasets, where the responses are always either `Yes` or `No`, another potential bias is whether `Yes` or `No` represents the positive option. Neither of these biases are present in the training data, as we have randomised the data during steering vector extraction such that the examples are split equally between the two (or four) choices. Despite this, we find these two biases are present (Figure 16) and are often highly predictive of the steerability, explaining a large part of the variance in per-example steerability (Figure 4).

This bias is different from the standard position or token bias exhibited by LLMs [40, 35], as it is a *steerability* bias: the model is *more steerable* towards the positive answer when it a particular position or token compared to the other position or token. The preferred token or position is different for each dataset; for example `corrigible-neutral-HHH` has a `B`-steerability bias, whereas `self-aware-lm` has an `A`-steerability bias (see Figure 16). This is problematic, as it is not fixable by simple dataset debiasing (which was already performed) or logit calibration adjustments [41] (as they effect propensity, not the change in propensity, i.e. steerability). Further, it implies that there may be other steerability biases present in models, determining when they are more or less steerable towards specific answers or behaviours. Indeed, there is still a high degree of unexplained variance present in many datasets in Figure 4.

**Some Behaviours are Un-Steerable.** We empirically observe that many behaviours turn out to be unsteerable, as measured by median steerability shown in Figure 1. One possible explanation is that the datasets we used were too small or low-quality. Other explanations include that unsteerable behaviours are not linearly represented in the model, or that they correspond to multiple separate behaviours within the model's ontology. In the latter case, it would be interesting to develop